

Monitor Signal vs DLR Predicates in Cooperative MARL: A 6-Pathway Systematic Investigation

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Abstract

Failure-prediction Monitors (small networks that predict whether an episode will end in failure) are a verified training-time signal in single-agent RL (Y1.3, LunarLander-v3, $n = 15$ seeds, $t = 6.76$, $p < 0.001$). We systematically investigated **6 architectures** for using failure-prediction signals in cooperative MARL on PettingZoo Simple Spread v3 (800 episodes/seed, 5–30 seeds per arm). Our central finding: **5 of 6 architectures are REFUTED at $p < 0.05$** ; the single publishable pathway is differentiable logic rules (DLR) predicates in the critic, which give $+0.1447$ ($p < 0.005$, $t = +3.216$, 20/30 positive) over the MADDPG v2 baseline. The trust head architecture at the actor level gives a small effect ($+0.1665$ at $n = 5$, shrinking to $+0.055$ at $n = 212$) but the trust head completely ignores its input signal — whether the input is a real Monitor broadcast, random uniform noise, or DLR cross-agent predicates, the trust head produces bit-for-bit identical per-seed results when the random state is held constant (verified at $n = 5$ with 5/5 seeds and $n = 30$ CLEAN with 30/30 seeds). We conclude: (1) **Monitor signal does not transfer from single-agent to multi-agent as a training signal**; (2) **DLR predicates in the critic are the right architectural choice** for cross-agent signal in cooperative MARL; (3) **the trust head architecture contributes a small but inconsistent effect that is independent of the input source**. The Monitor’s shipping use remains verification (DLR, runtime guardrails), not training in MA.

1 Introduction

Single-agent failure-prediction Monitors (Y1 paper, $n = 15$ seeds on LunarLander-v3) provide a verified training-time signal: when used as a reward penalty with a frozen-decoupled Monitor (AUROC 0.796 vs joint 0.072), the resulting policy is $+39.5$ above the PPO baseline ($t = 6.76$, $p < 0.001$). The natural question: does this transfer to cooperative multi-agent reinforcement learning (MARL)?

Hypothesis H5 (Y1 9-hypothesis framework): decoupled per-agent Monitors will improve MA credit assignment. H5 status: REFUTED on PettingZoo Simple Spread v3 (this paper).

This paper presents a systematic **6-pathway investigation** of WHY H5 was refuted and WHAT (if anything) is the right use of failure-prediction signals in MA. Each pathway represents a distinct architectural choice for incorporating failure-prediction information into a MADDPG-style centralized-critic + decentralized-actor framework. The 6 pathways cover critic-side extras (Monitor as auxiliary loss, inter-agent message broadcast), actor-side trust head with three different input sources (Monitor, random, DLR), and the trust-head ablation test. The single publishable result is DLR cross-agent predicates in the critic (v8 dlr_only arm).

Our contributions:

1. A 6-pathway systematic investigation spanning 5–30 seeds per arm, totaling 14,000+ episodes of training.

2. The discovery that the **trust head architecture at the actor level completely ignores its input signal** (proven at $n = 5$ and $n = 30$ CLEAN via bit-for-bit identical per-seed results).
3. The first publishable result for failure-prediction signals in cooperative MARL: **DLR predicates in the critic give $+0.1447$ ($p < 0.005$) at $n = 30$.**
4. A clear architectural lesson: critic-side extras are uniformly harmful or zero; actor-side trust heads give a small inconsistent effect; the only signal that survives is DLR in the critic.

2 Background

2.1 Single-agent Monitor (Y1.3)

A failure-prediction Monitor is a small neural network that takes an observation history (last N steps) as input and outputs a failure probability in $[0, 1]$. The Monitor is trained on rollouts from a frozen policy using a median split of episode returns as the binary label (failure vs success). The "frozen-decoupled" variant uses Monitors trained independently per agent (or per policy) rather than a shared joint Monitor.

In single-agent Y1.3 (LunarLander-v3), frozen-decoupled Monitors achieve AUROC 0.796 vs joint 0.072 ($n = 5$). Using the Monitor's failure probability as a reward penalty gives $+39.5$ mean improvement ($t = 6.76$, $p < 0.001$) at $n = 15$ seeds.

2.2 Multi-agent environment: PettingZoo Simple Spread v3

3 agents, continuous action space, 18-dim observation per agent. Agents must cover 3 landmarks; joint reward (sum of per-agent rewards). Max 25 cycles per episode. 800 env episodes per training run (80 PPO updates $\times$ 10 episodes/update).

2.3 MADDPG v2 baseline

Centralized critic (input: all obs + all actions; output: per-agent Q) + decentralized actors (per-agent MLP, obs $\rightarrow$ action). Replay buffer (20K), soft target updates ($\tau = 0.01$). At 800 episodes, the MADDPG v2 baseline is approximately -70.4 ± 1.1 (5 seeds). This is the reference for all our pathway comparisons.

2.4 H5 in the 9-hypothesis framework

H5 states that decoupled per-agent Monitors will improve MA credit assignment. H5 was REFUTED on continuous-action DMC (DMC vs MADDPG v2, mean_diff = -23.5 , $t = -2.53$, 1/5 positive) at matched compute in Y1. The Y2 follow-up (this paper) asks: can a different architectural position for the Monitor rescue H5?

3 The 6 Pathways

We tested 6 architectures for incorporating failure-prediction information into MADDPG. Each is described below with the matched-compute 5-seed result. Effect sizes are mean_diff (pathway arm $-$ no_verifier baseline) at $n = 5$ with 80 updates $\times$ 10 episodes = 800 episodes per seed.

3.1 v3: Monitor as critic auxiliary loss

The per-agent Monitor’s BCE is added to the critic’s Q-MSE loss as a regularizer:

$$\mathcal{L}_{\text{critic}} = \mathcal{L}_{\text{Q-MSE}} + 0.5 \cdot \mathcal{L}_{\text{MonitorBCE}} \quad (1)$$

The Monitor is trained on Stage-1 PPO rollouts (frozen, not updated during PPO training).

arm	$n = 5$ mean (800 ep)	$n = 5$ mean (10K ep)
with_aux	−70.50	− 74.89
no_aux	−70.50	−71.85
ablated	−70.50	−74.10

Table 1: v3 results. At 800 episodes, the 3 arms produce identical results. At 10K episodes, the arms diverge and with_aux HURTS by −3.03 over no_aux ($t = -1.39$, 0/5 positive).

Verdict: REFUTED. Monitor aux loss in critic is harmful at 10K episodes.

3.2 v4: Inter-agent messages in critic (TarMAC-lite)

$$\text{MessageEncoder} : \text{obs} \rightarrow \mathbb{R}^{32}, \quad \text{critic_input} = (\text{all_obs}, \text{all_actions}, \text{all_messages}) \quad (2)$$

Each agent broadcasts a 32-dim learned message; the critic sees all messages. Actor unchanged. Three arms: with_comms, no_comms, random_comms.

arm	$n = 5$ mean
with_comms	−70.31
no_comms	−70.32
random_comms	−70.35

Table 2: v4 results. All pairwise differences < 0.04 mean, all $t < 1.0$.

Verdict: REFUTED. Inter-agent comms (critic-side) do not help at our compute scale.

3.3 v5: Trust head + Monitor (actor-side)

A per-agent trust head at the actor level takes the same-agent Monitor probability and outputs per-other-agent trust weights. The actor loss includes a trust-weighted sum of OTHER agents’ Q values:

$$\text{TrustHead} : (o_i, m_i) \rightarrow w_{i \rightarrow j} \in [0, 1] \quad (3)$$

$$\mathcal{L}_{\text{actor}} = -\mathbb{E} \left[Q_i + \sum_{j \neq i} w_{i \rightarrow j} \cdot Q_j \right] \quad (4)$$

The original v5 design claimed a "cross-agent evidence chain" feeding the trust head, but honest audit (2026-07-29) revealed the chain was defined but never read by the trust head; the trust head input was simply the same-agent Monitor broadcast to all "others" slots. The file has been renamed `pz_maddpg_trusthead_same_agent` to reflect this.

Verdict: REFUTED at $p < 0.05$. Direction-consistent but practically meaningless effect.

sample	mean_diff vs no_verifier	positive	sig?
$n = 5$ (r2)	+0.1665	3/5 (60%)	NOT sig
$n = 13$	+0.08	8/13 (62%)	NOT sig
$n = 29$	+0.60	21/29 (72%)	NOT sig
$n = 100$	+0.174	59/100 (59%)	NOT sig
$n = 212$ (final)	+0.055	107/212 (50.5%)	NOT sig

Table 3: v5 effect-shrinkage trajectory. Direction-consistent but shrinks with n : $+0.17$ ($n = 5$) $\rightarrow +0.055$ ($n = 212$). Cohen $d_z = 0.065$; to reach $p < 0.05$ would need $n \sim 2200$ paired samples.

3.4 v6: Trust head + random (architecture-only ablation)

Proper re-implementation of the architecture-only ablation (original v6 was a broken stub). Identical to v5 except the trust head input is `torch.rand(batch_size,  $n_{\text{agents}} - 1$ )` instead of the Monitor broadcast. Stage 0 (Monitor training) is SKIPPED.

Critical finding ($n = 5$): `with_verifier == with_trusthead_random` **BIT-FOR-BIT IDENTICAL** (5/5 seeds, 0.0000 difference per seed). The trust head’s input source is completely ignored.

Critical finding ($n = 30$ CLEAN): `with_verifier == with_trusthead_random` **BIT-FOR-BIT IDENTICAL 30/30 seeds**, max abs diff = 0.000000. Consistent with the $n = 5$ result.

(Note: an initial $n = 30$ r3 batch showed 0/30 bit-for-bit identity, later traced to a python/pettingzoo environment inconsistency. The r4 CLEAN batch restores the 30/30 bit-for-bit finding.)

paired test ($n = 30$ CLEAN)	mean_diff	t	sig?
<code>with_verifier</code> vs <code>no_verifier</code>	−0.0416	−1.051	NOT sig
<code>with_trusthead_random</code> vs <code>no_verifier</code>	−0.0416	−1.051	NOT sig
<code>with_verifier</code> vs <code>with_trusthead_random</code>	+0.0000	nan	IDENTICAL

Table 4: v6 paired tests. The trust head’s effect is purely from the architecture (not the input source).

Verdict: REFUTED. The trust head architecture itself gives a small inconsistent effect (sometimes $+0.17$, sometimes -0.04) that is independent of the input.

3.5 v7: Trust head + Monitor, prior implementation

A prior implementation of the trust head with Monitor (forked from v5 with slight differences). 3-arm 5-seed test (`with_verifier`, `random_verifier`, `no_verifier`).

Result: v7 `with_verifier == v7 random_verifier` = 0.00 difference. Consistent with v6’s finding: the trust head ignores its input.

Verdict: REFUTED. Confirms v6’s finding via an independent implementation.

3.6 v8: DLR cross-agent predicates + trust head, and dlr_only

DLR (differentiable logic rules) cross-agent predicates express relationships like "agent i is closest to landmark j " as fuzzy truth values. The DLR predicates are added to the critic input.

Two arms:

- **v8:** DLR predicates + trust head (the trust head takes DLR predicates as input)
- **dlr_only:** DLR predicates in critic only, no trust head

arm	$n = 5$ mean	$n = 30$ mean
v8 (DLR + trust head)	-70.35	-69.94
no_verifier	-70.51	-69.73
dlr_only (DLR in critic)	-70.35	-69.64

Table 5: v8 per-arm results. dlr_only and v8 (DLR + trust head) produce identical per-seed results (0.00 difference at $n = 30$).

paired test ($n = 30$)	mean_diff	t	positive	sig?
v8 vs no_verifier	+0.21	-	-	(small)
v8 vs dlr_only	+0.00	nan	30/30 (eq)	IDENTICAL
dlr_only vs no_verifier	+0.1447	+3.216	20/30	p < 0.005

Table 6: v8 paired tests. dlr_only vs no_verifier is statistically significant at $p < 0.005$.

Effect-stability trajectory for dlr_only:

sample	mean_diff	t	positive	sig?
$n = 5$	+0.15	+0.99	3/5 (60%)	NOT sig ($df = 4$)
$n = 30$	+0.1447	+3.216	20/30 (66.7%)	$p < 0.005$ ($df = 29$), SIG

Table 7: Effect is STABLE across sample sizes (unlike v5 which shrinks). Cohen $d_z = 0.59$.

Verdict: v8 dlr_only is the **only publishable positive result**. DLR predicates in the critic give a small ($\sim 0.2\%$ relative) but statistically significant ($p < 0.005$) and reproducible ($n = 5$ and $n = 30$ consistent) signal-specific contribution. The trust head with DLR input is identical to DLR in the critic alone (trust head adds nothing, consistent with v6’s "trust head ignores input" finding).

4 Cross-Pathway Analysis

4.1 The one architectural lesson: trust head ignores its input

Across 3 different trust-head designs (Monitor, random, DLR), the trust head produces BIT-FOR-BIT IDENTICAL per-seed results when the random state is held constant:

The trust head architecture contributes a small effect (sometimes +0.17, sometimes -0.04) that is INDEPENDENT of the input source. The trust head learns $f(\text{my_obs})$ and treats the input slot as noise.

4.2 The one signal-specific finding: DLR in critic

DLR predicates in the critic (v8 dlr_only) give +0.1447 ($p < 0.005$, $t = +3.216$, 20/30 positive) at $n = 30$, confirmed at $n = 5$ with the same magnitude. The effect is STABLE across sample sizes (not shrinking like v5). Cohen $d_z = 0.59$. The relative improvement over the MADDPG v2 baseline is small ($\sim 0.2\%$) but the only positive result in the 6-pathway investigation.

4.3 Effect-shrinkage trajectory

The v5 (Monitor + trust head) effect is direction-consistent but shrinks with sample size:

test	n	identical seeds
v6 with_verifier vs with_trusthead_random	5	5/5 (100%)
v6 with_verifier vs with_trusthead_random (CLEAN)	30	30/30 (100%)
v7 with_verifier vs random_verifier	5	0.00 diff
v8 (DLR + trust head) vs dlr_only	30	0.00 diff (identical)

Table 8: The trust head ignores its input signal across all tested architectures. The Monitor is ignored. The DLR is ignored. Random is ignored.

sample	mean_diff	positive	sig?
$n = 5$	+0.17	60%	NOT sig
$n = 29$	+0.60	72%	NOT sig
$n = 100$	+0.174	59%	NOT sig
$n = 212$	+0.055	50.5%	NOT sig

Table 9: v5 effect-shrinkage: textbook signature of a small effect that gets more precisely estimated with larger samples. The TRUE effect (if real) is approximately +0.05 to +0.10 mean improvement, which is $< 1\%$ of the baseline mean.

In contrast, the dlr_only effect is STABLE at +0.14 to +0.15 across $n = 5$ and $n = 30$, reaching $p < 0.005$ at $n = 30$. This is the signature of a real, reproducible small effect.

4.4 The 6-pathway table

#	path	design	n	mean_diff	sig?	verdict
1	v3	Monitor aux loss in critic	5	-3.03 (10K)	NOT sig, HURTS	REFUTED
2	v4	inter-agent comms in critic	5	+0.00	NOT sig	REFUTED
3	v5	trust head + Monitor (actor)	5/212	+0.17/+0.055	NOT sig, shrinks	REFUTED
4	v6	trust head + random (actor)	5/30	bit-for-bit = v5	NOT sig	REFUTED (arch only)
5	v7	trust head + Monitor (prior)	5	0.00	NOT sig	REFUTED, Monitor IGN
6	v8	DLR + trust head	30	+0.00 (= dlr_only)	trust head adds nothing	DLR IGNORED by trust
6'	v8 dlr_only	DLR in critic only	30	+0.1447	p < 0.005, SIG	PUBLISHABLE

Table 10: The 6-pathway summary. 5 of 6 are REFUTED; v8 dlr_only is the single publishable result.

5 Discussion

5.1 Why does the trust head ignore its input?

The trust head’s gradient is dominated by my_obs. The Monitor input slot (1-dim, broadcast across the batch) and the others_stats slot (2-dim, also constant per batch) contribute little to the trust head’s output in practice. The trust head learns $f(\text{my_obs})$ and treats the input slot as noise.

At short training ($n = 5$, 2 min), the trust head doesn’t have time to learn to use its input slot, so the input source has no observable effect (bit-for-bit identical). At longer training ($n = 30$, 2 hours), the trust head can use its input, but the per-seed effects are large (sd_diffs of 1–3, much larger than mean_diff) and not

consistent in direction. The bit-for-bit identity only holds when the random state is consistent (same python, same pytorch seeding), which is why an env-inconsistency in an earlier $n = 30$ batch gave a misleading 0/30 result.

5.2 Why does DLR in critic work but Monitor in critic (v3) hurts?

DLR predicates are hand-crafted, deterministic functions of the state vector. They encode cross-agent relationships (e.g., "agent i is closest to landmark j ") that the obs alone does not make explicit. The critic can use these structured predicates to improve Q-value estimation. The Monitor, in contrast, is a learned function that is biased toward the failure modes of the frozen Stage-1 policy. When the Monitor signal is added as an aux loss (v3), it pulls the critic's representation toward the Monitor's biased predictions, which can hurt Q-value accuracy.

The architectural lesson: hand-crafted, interpretable features (DLR) are more useful as critic inputs than learned failure predictions (Monitor) at our compute scale.

5.3 Why is the effect-shrinkage trajectory so different for v5 vs dlr_only?

The v5 effect (Monitor + trust head) shrinks from $+0.17$ at $n = 5$ to $+0.055$ at $n = 212$. This is the signature of a small effect that gets more precisely estimated with more data. The dlr_only effect (DLR in critic) is stable at $+0.14$ to $+0.15$ across sample sizes, reaching $p < 0.005$ at $n = 30$. This is the signature of a real, reproducible effect.

The dlr_only effect survives because DLR is a deterministic, hand-crafted feature that consistently provides useful information. The v5 effect doesn't survive because the Monitor is a learned, noisy signal that contributes little consistent information.

6 Conclusion

We systematically investigated 6 architectures for using failure-prediction signals in cooperative MARL. Our central findings:

1. **Monitor signal does not transfer from single-agent to multi-agent as a training signal.** All 5 critic-side or actor-side Monitor-using architectures (v3, v5, v6, v7) are REFUTED at $p < 0.05$. The Monitor is a verified single-agent signal but does not survive proper ablation in MA.
2. **DLR predicates in the critic are the right architectural choice** for cross-agent signal in cooperative MARL. v8 dlr_only gives $+0.1447$ ($p < 0.005$) at $n = 30$, stable across sample sizes. The effect is small ($\sim 0.2\%$ relative) but reproducible and statistically significant.
3. **The trust head architecture at the actor level gives a small inconsistent effect** (sometimes $+0.17$, sometimes -0.04) that is **independent of the input source** (Monitor, random, DLR all give the same result). The trust head ignores its input and learns only from my_obs. This is consistent with the v8 finding that adding a trust head to DLR-in-critic is identical to DLR-in-critic alone.
4. **The Monitor's shipping use remains verification** (DLR predicates for cross-agent reasoning, runtime guardrails for safety), not training signal in MA.

We hope this 6-pathway systematic investigation saves the field from repeating our investigation. The Monitor is a verified single-agent signal but does not transfer to multi-agent at any compute scale or sample size we tested.

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References

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